

Genotype-dependent zinc partitioning and biochemical responses in tomato under excess zinc

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Highlights

- Zn strongly altered tomato biochemical traits without major growth effects.
- ST-P accumulated more Zn in leaves than Heinz under excess Zn.
- Heinz retained more Zn in roots at the highest Zn treatment.
- ST-P showed higher constitutive guaiacol peroxidase activity.
- Zn partitioning was strongly associated with chlorophyll loss.

Abstract

Zinc (Zn) is essential but becomes phytotoxic when cellular homeostasis is exceeded. We compared Zn partitioning and biochemical responses of two tomato genotypes, Heinz and Saint-Pierre (ST-P), exposed to 0, 100 and 250 mg kg⁻¹ ZnSO₄·7H₂O in a greenhouse experiment lasting 55 days. The randomized complete block design comprised two genotypes, three Zn treatments and three blocks; three plants within each experimental unit were treated as subsamples. Growth, relative water content, soluble proteins, chlorophylls, carotenoids, tissue Zn and guaiacol peroxidase (POD) activity were measured. Zn had limited effects on morphology but strongly altered biochemical traits. Leaf soluble protein increased by 34.0% at 100 mg kg⁻¹ and decreased by 30.3% at 250 mg kg⁻¹, whereas root protein remained elevated at the higher dose. Chlorophyll *a* and *b* decreased by 49.6% and 62.8%, respectively, at 250 mg kg⁻¹. POD activity increased with Zn exposure and was consistently higher in ST-P; basal activity was 35.5% higher in ST-P than in Heinz. At 250 mg kg⁻¹, Heinz retained more Zn in roots, whereas ST-P accumulated more Zn in leaves, giving translocation factors of 0.74 and 1.25, respectively. Experimental-unit analyses showed strong negative associations between tissue Zn and total chlorophyll. The results indicate genotype-dependent Zn partitioning and antioxidant responses.

The higher ST-P translocation factor is consistent with greater shoot partitioning, but intracellular Zn sequestration and transporter activity were not directly assessed.

Keywords: *Solanum lycopersicum*; zinc toxicity; zinc partitioning; translocation factor; peroxidase; oxidative stress

1. Introduction

Zinc (Zn) is an essential micronutrient required for plant growth and metabolism, contributing to enzyme activity, protein structure, transcriptional regulation and cellular homeostasis. However, excessive Zn availability disrupts the balance between nutritional requirement and metal toxicity and can impair root development, photosynthetic function, mineral nutrition and cellular redox homeostasis [1–3]. The physiological response to Zn depends not only on total soil Zn but also on its chemical availability, soil properties, uptake processes and the ability of plants to distribute and detoxify Zn within their tissues [1].

Tomato (*Solanum lycopersicum* L.) is an economically important crop with substantial genetic diversity. Genotypes can differ in their responses to abiotic stress through differences in root uptake, tissue partitioning, antioxidant capacity and metabolic adjustment. Comparative physiological studies are therefore useful for identifying traits associated with Zn tolerance and for distinguishing restricted metal uptake from internal tolerance mechanisms.

At excessive concentrations, Zn can inhibit root development, disturb mineral homeostasis, impair photosynthetic pigments and alter cellular redox balance [2,4]. Plants respond through coordinated mechanisms that include modification of root uptake, binding of metals to cell-wall components, chelation by organic ligands, intracellular sequestration and antioxidant defense [3,5]. Several transporter families participate in Zn homeostasis, including ZIP, HMA and MTP proteins [1,6]. In tomato, members of the MTP family respond differentially to heavy-metal exposure, indicating that these transporters may contribute to tissue-specific Zn homeostasis [6].

Root cell walls may also influence Zn partitioning. Pectins, hemicelluloses and other negatively charged cell-wall components can bind metal ions in the apoplast and thereby influence their movement into living cells and toward the vascular system [7]. Consequently, differences in root Zn retention among genotypes may arise from multiple processes, including apoplastic binding, membrane transport and vascular loading rather than from a single transport mechanism.

The translocation factor (TF), calculated as the ratio of Zn concentration in leaves to that in roots, provides a simple descriptor of tissue partitioning:

$$TF = [Zn]_{leaf}/[Zn]_{root}$$

A TF below 1 indicates greater Zn concentration in roots than leaves, whereas a value above 1 indicates greater relative partitioning to leaves. TF is useful for describing Zn distribution but does not by itself demonstrate transporter activity or intracellular compartmentation.

Zn responses may also be nonlinear. Moderate exposure can stimulate some physiological processes, whereas excessive exposure can become inhibitory. Such biphasic responses may be compatible with hormesis, although a rigorous demonstration requires an appropriate dose-

response design over a sufficiently broad concentration range [8]. Consequently, a biphasic response observed at only three concentrations should be interpreted cautiously.

Antioxidant enzymes are important components of the response to metal-induced oxidative imbalance. Peroxidases participate in H_2O_2 metabolism and can respond strongly to metal exposure [9–11]. However, peroxidase activity represents only one component of the antioxidant system, and its interpretation is most informative when considered together with Zn accumulation and other physiological responses.

The present study compared the industrial tomato cultivar Heinz and the traditional Saint-Pierre genotype (ST-P) under soil-based Zn exposure. We hypothesized that the genotypes would differ in Zn partitioning and antioxidant responses. Specifically, we tested whether ST-P would show greater shoot Zn partitioning and higher basal POD activity, whereas Heinz would retain a greater proportion of Zn in roots.

2. Materials and methods

2.1. Plant material and experimental conditions

Two tomato genotypes were used: Heinz (Hnz), an industrial determinate-growth cultivar (cv. ‘Heinz 1350’; HeinzSeed, Pittsburgh, PA, USA; lot #HNZ-2022-OR1), and Saint-Pierre (ST-P), a traditional indeterminate-growth cultivar (Seed Savers Exchange, Decorah, IA, USA; accession #SSE-TOM-047).

ST-P is recognized as a Mediterranean tomato landrace in studies of tomato genetic diversity [12]. This reference is used here to establish the genetic and horticultural background of ST-P and not as direct evidence of Zn tolerance. The comparative Zn-response phenotype reported in this study is based on the present experiment.

Seeds were surface-sterilized in 5% sodium hypochlorite for 10 min, rinsed three times with sterile distilled water, and germinated in sterilized peat/sand substrate (1:1, v/v). Seedlings at the 3–5 true-leaf stage, approximately 21 days after germination, were transplanted into 1-kg pots containing the experimental substrate. Plants were maintained under greenhouse conditions with a natural photoperiod and a temperature range of approximately 15–20 °C.

2.2. Experimental design and Zn treatments

A randomized complete block design was used with genotype and Zn treatment as fixed factors. The experiment comprised two genotypes (Heinz and ST-P) and three Zn treatments: Zn0, Zn100 and Zn250.

$\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ (purity $\geq 99\%$; Sigma-Aldrich, St. Louis, MO, USA) was applied at nominal concentrations of 0, 100 and 250 mg kg^{-1} substrate. Each genotype $\times$ treatment combination was represented in three independent blocks. Each experimental unit contained three plants, giving 54 plants in total (2 genotypes $\times$ 3 treatments $\times$ 3 blocks $\times$ 3 plants). The experiment lasted 55 days after transplanting and ended at the full-flowering stage.

The experimental unit was the block-level genotype $\times$ treatment combination. The three plants within an experimental unit were considered subsamples and were not treated as independent biological replicates.

2.3. Morphological measurements

At harvest, stem length and root length were measured. Fresh weights of stems, leaves and roots were determined using an analytical balance with a precision of 0.0001 g. Dry weight was determined after oven-drying at 70 °C for 48 h.

Relative water content (RWC) was calculated according to:

$$\text{RWC (\%)} = [(\text{FW} - \text{DW})/(\text{TW} - \text{DW})] \times 100$$

where FW is fresh weight, TW is turgid weight after 24 h in distilled water and DW is dry weight [13].

2.4. Biochemical analyses

Leaf and root tissues were collected at the end of the experiment, immediately placed on ice and processed at 4 °C.

Soluble proteins were quantified using the Bradford method with bovine serum albumin as the standard and absorbance measured at 595 nm [14].

Chlorophylls and carotenoids were extracted using 80% acetone (v/v) following Lichtenthaler and Wellburn [15]. Absorbance was measured at 665, 652 and 470 nm.

2.5. Zn determination by flame atomic absorption spectrometry

Zn concentrations in leaf and root tissues were determined by flame atomic absorption spectrometry (FAAS) at 213.9 nm following acid digestion with HNO₃ (65%) and H₂O₂ (30%) at 120 °C. Measurements were performed using an AAnalyst 400 spectrometer (PerkinElmer, Waltham, MA, USA).

Quality assurance included procedural blanks consisting of the HNO₃/H₂O₂ digestion matrix, a Zn calibration range of 0.05–5.0 mg L⁻¹, and a calibration coefficient of determination of R² = 0.9997. Spike recovery was 97.3 $\pm$ 3.1% (n = 6). The limit of detection and limit of quantification were 0.02 and 0.07 mg kg⁻¹, respectively. NIST SRM 1573a tomato leaves were used as the certified reference material; measured Zn concentration was 30.8 $\pm$ 1.2 mg kg⁻¹ compared with the certified value of 30.9 $\pm$ 0.7 mg kg⁻¹.

The translocation factor was calculated as:

$$\text{TF} = [\text{Zn}]_{\text{leaf}}/[\text{Zn}]_{\text{root}}$$

All Zn concentrations were expressed on a dry-weight basis.

2.6. Peroxidase activity

Guaiacol peroxidase (POD; EC 1.11.1.7) activity was determined spectrophotometrically following Chance and Maehly [16]. The 3-mL reaction mixture contained 2.7 mL phosphate buffer (0.1 M, pH 7.0), 100 μ L guaiacol (1%, v/v), 100 μ L H₂O₂ (0.3%, v/v) and 100 μ L crude enzyme extract.

The increase in tetraguaiacol absorbance was monitored at 470 nm for 120 s. Activity was calculated from the initial linear interval between 0 and 60 s using the molar extinction coefficient of tetraguaiacol ($\epsilon = 26.6 \text{ mM}^{-1} \text{ cm}^{-1}$). One unit of POD activity was defined as the amount of enzyme catalyzing the oxidation of 1 μ mol guaiacol min^{-1} under the assay conditions.

2.7. Statistical analysis

Statistical analyses were performed using SPSS version 26.0 (IBM Corp., Armonk, NY, USA) and R version 4.3.1 (R Foundation for Statistical Computing, Vienna, Austria).

Data were analyzed using two-way analysis of variance to test genotype, Zn treatment and genotype $\times$ treatment effects. The randomized complete block structure was retained in the analysis. Tukey's HSD test was used for multiple comparisons at $\alpha = 0.05$. Effect sizes were expressed as partial eta-squared (η^2).

Because the three plants within each experimental unit were subsamples, they were not treated as independent biological replicates. Correlation analyses were therefore conducted using experimental-unit means. Genotype-specific analyses were based on the nine experimental units corresponding to three Zn treatments $\times$ three blocks for each genotype.

As a sensitivity analysis, linear mixed-effects models incorporating experimental unit as a random effect were also fitted. The resulting interpretation was consistent with the primary analyses.

3. Results

3.1. Morphological responses

Zn exposure produced limited effects on the measured morphological traits. No significant genotype, treatment or genotype $\times$ treatment effects were detected for stem length, root length or organ fresh weights ($p > 0.05$).

Dry weight differed significantly between genotypes ($F = 5.114$, $p = 0.043$, $\eta^2 = 0.266$), with higher values in Heinz ($46.57 \pm 3.56 \text{ mg}$) than ST-P ($43.60 \pm 2.15 \text{ mg}$). RWC remained between 86.31% and 92.57% across treatments.

Thus, the Zn treatments produced relatively limited morphological effects at the harvest stage despite pronounced changes in several biochemical variables.

3.2. Soluble protein responses

Leaf soluble protein concentration was significantly affected by genotype ($F = 475.695$, $p < 0.001$) and Zn treatment ($F = 685.909$, $p < 0.001$, $\eta^2 = 0.723$). Protein concentration increased by 34.0% at Zn100 relative to the control, from 20.83 ± 3.26 to 27.91 ± 3.51 mg g⁻¹ FW, and decreased by 30.3% at Zn250 to 14.52 ± 3.66 mg g⁻¹ FW.

Root soluble protein concentration was significantly affected by genotype ($F = 482.783$, $p < 0.001$), treatment ($F = 439.179$, $p < 0.001$, $\eta^2 = 0.599$) and their interaction ($F = 28.612$, $p < 0.001$). Root protein concentration increased by 62.9% at Zn100 and remained 23.8% above the control at Zn250.

The increase followed by decline in leaf protein represents a biphasic response consistent with hormesis, although the present three-dose experiment does not establish a complete hormetic dose-response relationship.

3.3. Zn accumulation and tissue partitioning

Foliar Zn concentration was strongly affected by Zn treatment ($F = 2755.143$, $p < 0.001$, $\eta^2 = 0.961$).

In Heinz, leaf Zn concentration increased from 25.85 ± 5.30 mg kg⁻¹ DW in the control to 252.40 ± 7.14 mg kg⁻¹ DW at Zn250. In ST-P, leaf Zn increased from 20.07 ± 1.64 to 310.70 ± 12.38 mg kg⁻¹ DW.

At Zn250, root Zn concentration was higher in Heinz (339.09 ± 41.99 mg kg⁻¹ DW) than in ST-P (249.11 ± 27.72 mg kg⁻¹ DW).

The resulting TF values were 0.74 for Heinz and 1.25 for ST-P. Thus, Heinz showed greater relative Zn retention in roots, whereas ST-P showed greater relative partitioning of Zn to leaves.

Because only total tissue Zn was measured, these differences demonstrate contrasting tissue partitioning but do not directly establish intracellular compartmentation or transporter activity.

3.4. Photosynthetic pigments

Zn treatment significantly affected chlorophyll *a* ($F = 97.377$, $p < 0.001$, $\eta^2 = 0.795$) and chlorophyll *b* ($F = 28.213$, $p < 0.001$, $\eta^2 = 0.532$), whereas genotype effects were not significant.

Chlorophyll *a* decreased from 19.13 ± 2.92 µg mL⁻¹ in the control to 13.67 ± 1.65 µg mL⁻¹ at Zn100 and 9.65 ± 1.06 µg mL⁻¹ at Zn250. These values corresponded to reductions of 28.6% and 49.6%, respectively.

Chlorophyll *b* decreased from 8.36 ± 2.76 µg mL⁻¹ in the control to 5.10 ± 2.18 and 3.11 ± 0.86 µg mL⁻¹ at Zn100 and Zn250, corresponding to reductions of 39.1% and 62.8%.

Carotenoid concentration was not significantly affected by treatment ($F = 1.780$, $p = 0.180$).

3.5. Peroxidase activity

POD activity was significantly affected by genotype ($F = 72.887$, $p < 0.001$), Zn treatment ($F = 2815.231$, $p < 0.001$, $\eta^2 = 0.946$), and their interaction ($F = 99.197$, $p < 0.001$).

In Heinz, POD activity increased from 0.077 U mL^{-1} in the control to 0.101 U mL^{-1} at Zn100 and 0.116 U mL^{-1} at Zn250. In ST-P, the corresponding values were 0.105, 0.132 and 0.153 U mL^{-1} .

ST-P therefore exhibited 35.5% higher basal POD activity than Heinz. At Zn250, POD-specific activity increased by 152.4% in Heinz and 95.3% in ST-P relative to the respective controls.

These results indicate a strong Zn-associated induction of POD activity and a higher constitutive POD activity in ST-P.

3.6. Associations among Zn accumulation and biochemical variables

Correlation analyses were performed using experimental-unit means rather than individual plants.

The analysis showed strong negative associations between leaf Zn and total chlorophyll ($r = -0.847$, $p = 0.004$) and between root Zn and total chlorophyll ($r = -0.823$, $p < 0.01$). Root Zn was positively associated with POD activity ($r = 0.684$, $p = 0.042$), whereas leaf soluble protein was negatively associated with POD activity ($r = -0.712$, $p = 0.031$).

Genotype-specific analyses showed similar negative associations between leaf Zn and total chlorophyll in Heinz ($r = -0.891$, $p = 0.012$) and ST-P ($r = -0.823$, $p = 0.023$).

These associations indicate coordinated changes in Zn accumulation, pigment status and antioxidant activity but do not establish causal relationships.

4. Discussion

4.1. Zn exposure produced stronger biochemical than morphological responses

The relatively limited effects of Zn on morphology contrasted with the pronounced changes in soluble proteins, chlorophylls, tissue Zn and POD activity. This pattern indicates that the concentrations used in the experiment were sufficient to perturb plant biochemical status without producing severe visible growth inhibition at the harvest stage.

The maintenance of RWC also indicates that the biochemical responses were not simply consequences of severe water deficit. The results therefore support the use of complementary biochemical and physiological measurements when evaluating Zn stress.

However, the experiment was conducted at a single harvest time. The present results consequently cannot establish whether biochemical variables respond earlier than morphological

traits. The appropriate conclusion is that biochemical variables were **more strongly affected than the measured morphological traits at the sampling stage**, rather than that they constitute proven early-warning indicators.

4.2. Biphasic protein response under Zn exposure

The increase in leaf soluble protein at Zn100 followed by a decrease at Zn250 represents a biphasic response consistent with hormesis. Low or moderate Zn exposure can stimulate acclimatory processes, whereas excessive Zn can interfere with cellular metabolism and protein homeostasis [1,8].

Several mechanisms could contribute to the observed pattern, including altered protein synthesis, stress-associated proteins, metal-binding compounds and oxidative damage. However, total soluble protein was measured in the present study, and individual protein classes were not characterized. Therefore, the increase cannot be specifically attributed to metallothioneins, phytochelatins, heat-shock proteins or other defined stress proteins.

The stronger reduction in leaf protein at Zn250 in Heinz may indicate greater biochemical sensitivity of this genotype at the highest treatment. This remains a physiological interpretation rather than a molecular mechanism.

4.3. Genotype-dependent Zn partitioning

The clearest difference between the genotypes concerned tissue Zn distribution. Heinz accumulated more Zn in roots at Zn250, whereas ST-P accumulated more Zn in leaves. The corresponding TF values were 0.74 and 1.25, respectively.

These results demonstrate contrasting Zn partitioning strategies. Greater root retention can restrict Zn movement toward leaves, whereas greater shoot partitioning can expose leaves to higher Zn concentrations. Whether either strategy constitutes greater tolerance depends on the capacity of the tissue receiving Zn to maintain cellular homeostasis.

Several processes may contribute to the observed difference. Root cell-wall components, particularly pectins and hemicelluloses, can bind metal ions and influence their movement through the apoplast [7]. Membrane transport systems subsequently regulate Zn uptake, efflux and movement toward the vascular tissues. HMA proteins contribute to metal transport in plants, while MTP/CDF proteins participate in intracellular metal homeostasis [1,6].

Tomato MTP genes exhibit differential responses to heavy-metal exposure, providing a plausible molecular basis for genotype-dependent Zn partitioning [6]. Nevertheless, the present experiment did not quantify transporter transcript abundance, transporter proteins, xylem Zn loading or cell-wall Zn binding.

Similarly, the TF of 1.25 in ST-P should not be interpreted as direct evidence of vacuolar Zn sequestration. Intracellular compartmentation would require subcellular localization or fractionation experiments. The present data therefore support the more conservative interpretation that ST-P has **greater shoot Zn partitioning**, whereas Heinz shows **greater root Zn retention**.

4.4. Chlorophyll responses indicate substantial physiological perturbation

The pronounced reductions in chlorophyll *a* and *b* at Zn250 indicate strong perturbation of the photosynthetic apparatus or chlorophyll metabolism. Excess Zn can affect photosynthesis through several mechanisms, including disturbance of mineral nutrition, interference with chlorophyll metabolism and oxidative damage [1,2].

The present study did not measure photosynthetic rate, chlorophyll fluorescence, mineral concentrations other than Zn, or chlorophyll biosynthetic enzymes. Consequently, the molecular basis of chlorophyll loss cannot be resolved from these data.

Interestingly, carotenoid concentration did not differ significantly among treatments. This result indicates that Zn affected the measured chlorophyll pools more strongly than the carotenoid pool under the conditions tested, but it should not be interpreted as direct evidence of preserved photoprotection.

The absence of a significant genotype effect on chlorophyll suggests that the contrasting Zn responses of Heinz and ST-P cannot be explained simply by differential maintenance of the measured pigment pools.

4.5. Constitutive POD activity distinguishes the genotypes

POD activity increased progressively with Zn exposure and showed a strong treatment effect. ST-P also had substantially higher basal POD activity than Heinz.

This pattern is compatible with differences in antioxidant capacity. Peroxidases contribute to H₂O₂ metabolism and cell-wall-associated redox processes and are frequently responsive to metal stress [9–11]. The higher basal POD activity in ST-P may therefore represent a constitutive biochemical trait associated with its response to Zn.

The stronger relative increase in POD-specific activity in Heinz at Zn250 should be interpreted cautiously. A large inducible response does not necessarily indicate greater tolerance; it may instead reflect greater stress intensity in the tissue. Conversely, the higher constitutive activity of ST-P may provide greater basal capacity for peroxide metabolism.

The present experiment measured POD only. It did not determine H₂O₂, superoxide, lipid peroxidation, catalase, superoxide dismutase, ascorbate peroxidase or glutathione. Therefore, the data support an association between POD activity and Zn exposure but do not establish the complete antioxidant mechanism responsible for genotype-dependent tolerance.

4.6. Relationships between Zn accumulation and biochemical responses

The strong negative associations between tissue Zn and total chlorophyll indicate that greater Zn accumulation was associated with stronger pigment loss under the experimental conditions.

However, correlation does not demonstrate causality. Treatment level itself generated the major Zn concentration gradient and could therefore contribute to the observed correlations.

Accordingly, tissue Zn should be described as being **strongly associated with chlorophyll loss**, rather than as the primary causal driver of photosynthetic damage.

The positive relationship between root Zn and POD activity is consistent with activation of antioxidant responses under greater Zn exposure. Again, this is an association rather than evidence that Zn directly induced POD through a specific signaling pathway.

The genotype-specific correlations provide additional evidence that the Zn–chlorophyll relationship was not restricted to one genotype. Nevertheless, the limited number of experimental units means that these associations should be considered exploratory and should be validated in larger experiments.

4.7. Physiological interpretation of contrasting Zn responses

Taken together, the results support a model in which Heinz and ST-P differ primarily in Zn partitioning and basal antioxidant status.

Heinz retained more Zn in roots at the highest treatment, while ST-P transferred a greater proportion of Zn to leaves. ST-P also had higher basal POD activity. These characteristics are consistent with two different physiological strategies: stronger root retention in Heinz and greater shoot partitioning combined with a comparatively high constitutive antioxidant capacity in ST-P.

The present data do not establish whether the higher shoot Zn concentration in ST-P is beneficial or detrimental in terms of whole-plant fitness. Greater shoot accumulation may represent tolerance if Zn can be safely immobilized, but it may also increase the risk of Zn accumulation in edible tissues. Conversely, root retention may protect shoots from Zn exposure but increase metal loading in roots.

This distinction is particularly important for crop improvement. A high TF should not automatically be considered a desirable trait for food crops. Its value depends on the intended production environment, edible-tissue Zn concentration, yield stability and food-safety considerations.

4.8. Implications for future physiological studies

The contrasting phenotypes provide a useful basis for mechanistic follow-up experiments. In particular, comparison of the two genotypes using subcellular Zn localization, root cell-wall fractionation, xylem sap analysis and expression analysis of candidate Zn transporters could determine whether the observed TF difference reflects differences in apoplastic retention, vascular loading or intracellular sequestration.

Combining these approaches with measurements of ROS, lipid peroxidation and several antioxidant enzymes would also establish whether the higher constitutive POD activity of ST-P corresponds to a broader antioxidant phenotype.

Plant–microbe interactions represent an additional direction for future research because rhizosphere microorganisms can modify metal availability and plant stress responses. Such interactions were not evaluated in the present study and therefore cannot be invoked as an explanation for the observed genotype differences.

5. Conclusions

The two tomato genotypes exhibited contrasting physiological responses to excess Zn. Morphological traits were comparatively stable, whereas soluble proteins, chlorophylls, tissue Zn concentrations and POD activity responded strongly.

Heinz accumulated more Zn in roots at the highest Zn treatment and showed a TF below 1, whereas ST-P accumulated more Zn in leaves and showed a TF above 1. ST-P also exhibited higher constitutive POD activity.

The protein response at moderate and high Zn concentrations was biphasic and consistent with hormesis, although the present three-dose experiment does not establish a complete hormetic dose-response relationship. The higher TF of ST-P indicates greater shoot Zn partitioning but does not by itself demonstrate vacuolar sequestration or transporter activity.

Overall, the results identify genotype-dependent Zn partitioning and constitutive POD activity as physiological traits worthy of further investigation. Future studies combining tissue-level Zn measurements with subcellular localization, transporter analysis and broader oxidative-stress profiling should clarify the mechanisms underlying the contrasting responses.

What's next

The principal next step is to determine whether the contrasting Zn-partitioning patterns arise from differences in root apoplastic retention, vascular loading or intracellular sequestration. Integrating Zn localization with transporter expression and antioxidant profiling should help distinguish these mechanisms and determine whether the ST-P phenotype represents genuine physiological tolerance or simply greater shoot Zn accumulation.

Credit authorship contribution statement

Allouche A.: Conceptualization, Investigation, Methodology, Formal analysis, Data curation, Writing – original draft.

Arbaoui M.: Methodology, Investigation, Supervision, Writing – review and editing.

Touati A.: Investigation, Methodology, Data interpretation, Writing – review and editing.

Yahia N.: Conceptualization, Supervision, Resources, Writing – review and editing.

All authors approved the final version of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Ethics statement

This study did not involve human participants or experimental animals.

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References

1. Stanton C, Sanders D, Krämer U, Podar D. Zinc in plants: integrating homeostasis and biofortification. *Mol Plant*. 2022;15:65–85. <https://doi.org/10.1016/j.molp.2021.12.008>.
2. Broadley MR, White PJ, Hammond JP, Zelko I, Lux A. Zinc in plants. *New Phytol*. 2007;173:677–702. <https://doi.org/10.1111/j.1469-8137.2007.01996.x>.
3. Hall JL. Cellular mechanisms for heavy metal detoxification and tolerance. *J Exp Bot*. 2002;53:1–11. <https://doi.org/10.1093/jexbot/53.366.1>.
4. Marschner H. *Mineral Nutrition of Higher Plants*. 2nd ed. London: Academic Press; 1995.
5. Cobbett C, Goldsbrough P. Phytochelatins and metallothioneins: roles in heavy metal detoxification and homeostasis. *Annu Rev Plant Biol*. 2002;53:159–182. <https://doi.org/10.1146/annurev.arplant.53.100301.135154>.
6. El-Sappah AH, Elrys AS, Desoky EM, Zhao X, Wang B, El-Sappah HH, et al. Comprehensive genome wide identification and expression analysis of MTP gene family in tomato (*Solanum lycopersicum*) under multiple heavy metal stress. *Saudi J Biol Sci*. 2021;28:6946–6956. <https://doi.org/10.1016/j.sjbs.2021.07.073>.

7. Krzesłowska M. The cell wall in plant cell response to trace metals: polysaccharide remodeling and its role in defense strategy. *Acta Physiol Plant*. 2011;33:35–51. <https://doi.org/10.1007/s11738-010-0521-8>.
8. Calabrese EJ, Baldwin LA. Hormesis: U-shaped dose responses and their centrality in toxicology. *Trends Pharmacol Sci*. 2001;22:285–291. [https://doi.org/10.1016/S0165-6147\(00\)01719-3](https://doi.org/10.1016/S0165-6147(00)01719-3).
9. Mittler R. Oxidative stress, antioxidants and stress tolerance. *Trends Plant Sci*. 2002;7:405–410. [https://doi.org/10.1016/S1360-1385\(02\)02312-9](https://doi.org/10.1016/S1360-1385(02)02312-9).
10. Foyer CH, Noctor G. Redox homeostasis and antioxidant signaling: a metabolic interface between stress perception and physiological responses. *Plant Cell*. 2005;17:1866–1875. <https://doi.org/10.1105/tpc.105.033589>.
11. Cuypers A, Hendrix S, Amaral dos Reis R, De Smet S, Deckers J, Gielen H, et al. Hydrogen peroxide, signaling in disguise during metal phytotoxicity. *Front Plant Sci*. 2016;7:470. <https://doi.org/10.3389/fpls.2016.00470>.
12. Terzopoulos PJ, Bebeli PJ. Genetic diversity analysis of Mediterranean tomato landraces using 10 highly polymorphic SSR markers. *Sci Hortic*. 2008;116:272–277. <https://doi.org/10.1016/j.scienta.2008.01.003>.
13. Barrs HD, Weatherley PE. A re-examination of the relative turgidity technique for estimating water deficits in leaves. *Aust J Biol Sci*. 1962;15:413–428. <https://doi.org/10.1071/BI9620413>.
14. Bradford MM. A rapid and sensitive method for the quantitation of microgram quantities of protein utilizing the principle of protein-dye binding. *Anal Biochem*. 1976;72:248–254. [https://doi.org/10.1016/0003-2697\(76\)90527-3](https://doi.org/10.1016/0003-2697(76)90527-3).
15. Lichtenthaler HK, Wellburn AR. Determinations of total carotenoids and chlorophylls *a* and *b* of leaf extracts in different solvents. *Biochem Soc Trans*. 1983;11:591–592. <https://doi.org/10.1042/bst0110591>.
16. Chance B, Maehly AC. Assay of catalases and peroxidases. *Methods Enzymol*. 1955;2:764–775. [https://doi.org/10.1016/S0076-6879\(55\)02300-8](https://doi.org/10.1016/S0076-6879(55)02300-8).
17. Hussain D, Haydon MJ, Wang Y, Wong E, Sherson SM, Young J, et al. P-type ATPase heavy metal transporters with roles in essential zinc homeostasis in *Arabidopsis*. *Plant Cell*. 2004;16:1327–1339. <https://doi.org/10.1105/tpc.020487>.
18. Verbruggen N, Hermans C, Schat H. Molecular mechanisms of metal hyperaccumulation in plants. *New Phytol*. 2009;181:759–776. <https://doi.org/10.1111/j.1469-8137.2008.02748.x>.
19. Clemens S, Palmgren MG, Kramer U. A long way ahead: understanding and engineering plant metal accumulation. *Trends Plant Sci*. 2002;7:309–315. [https://doi.org/10.1016/S1360-1385\(02\)02295-1](https://doi.org/10.1016/S1360-1385(02)02295-1).

20. Gasic K, Korban SS. Expression of Arabidopsis phytochelatase in Indian mustard (*Brassica juncea*) plants enhances tolerance for Cd and Zn. *Planta*. 2007;225:1277–1285. <https://doi.org/10.1007/s00425-006-0421-y>.
21. Cuypers A, Karen S, Jos R, Kelly H, Tony R, Nele H, et al. The cellular redox state as a modulator in cadmium and copper responses in *Arabidopsis thaliana* seedlings. *J Plant Physiol*. 2011;168:309–316. <https://doi.org/10.1016/j.jplph.2010.07.010>.
22. Dat J, Vandenabeele S, Vranová E, Van Montagu M, Inzé D, Van Breusegem F. Dual action of the active oxygen species during plant stress responses. *Cell Mol Life Sci*. 2000;57:779–795. <https://doi.org/10.1007/s000180050041>.
23. Sharma SS, Dietz KJ. The significance of amino acids and amino acid-derived molecules in plant responses and adaptation to heavy metal stress. *J Exp Bot*. 2006;57:711–726. <https://doi.org/10.1093/jxb/erj073>.
24. Fontes RLF, Cox FR. Zinc toxicity in soybean grown at high iron concentration in nutrient solution. *J Plant Nutr*. 1998;21:1723–1730. <https://doi.org/10.1080/01904169809365517>.
25. Kupper H, Kupper F, Spiller M. Environmental relevance of heavy metal-substituted chlorophylls using the example of water plants. *J Exp Bot*. 1996;47:259–266. <https://doi.org/10.1093/jxb/47.2.259>.
26. Kupper H, Zhao FJ, McGrath SP. Cellular compartmentation of zinc in leaves of the hyperaccumulator *Thlaspi caerulescens*. *Plant Physiol*. 1999;119:305–311. <https://doi.org/10.1104/pp.119.1.305>.
27. Murtic S, Zahirovic C, Civic H, Karic L, Jurkovic J. Uptake of heavy metals by tomato plants and their distribution inside the plant. *Agric For*. 2018;64:251–261. <https://doi.org/10.17707/AgricultForest.64.4.25>.
28. Tabassam Q, Ahmad MSA, Alvi AK, Awais M, Kaushik P, El-Sheikh MA. Accumulation of different metals in tomato fruits irrigated with wastewater. *Appl Sci*. 2023;13:9711. <https://doi.org/10.3390/app13179711>.
29. Sanad H, Moussadek R, Spaccini R, Paradiso R, Oueld Lhaj M, Zouahri A, et al. Trace metal accumulation in horticulture production systems of Mediterranean agro-ecosystems: origins, impacts on soil health, water resources, and plant uptake with sustainable mitigation strategies. *Front Sustain Food Syst*. 2026;10:1803164. <https://doi.org/10.3389/fsufs.2026.1803164>.
30. Alloway BJ. *Heavy Metals in Soils: Trace Metals and Metalloids in Soils and their Bioavailability*. 3rd ed. Dordrecht: Springer; 2013. <https://doi.org/10.1007/978-94-007-4470-7>.
31. Kabata-Pendias A, Mukherjee AB. *Trace Elements from Soil to Human*. Berlin: Springer; 2007. <https://doi.org/10.1007/978-3-540-32714-1>.

32. USEPA. Method 3050B: Acid Digestion of Sediments, Sludges, and Soils. Washington, DC: United States Environmental Protection Agency; 1996.